

TUSHAR SAURABH

AUTOMATION • OBSERVABILITY • AGENTIC AI

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PROFESSIONAL SUMMARY

Hands-on engineer focused on solving problems across production engineering, observability, automation, software engineering, and emerging AI-assisted workflows. I build software that removes repetitive work, improves troubleshooting, and creates longer-term engineering solutions rather than treating each operational problem in isolation.

My experience spans production diagnostics, application performance and reliability, engineering automation, developer tooling, and application modernization. I am currently exploring where agentic AI can improve these workflows while continuing to use deterministic approaches where correctness, consistency, and repeatability matter.

CORE CAPABILITIES

Production Engineering & Troubleshooting: Production diagnostics • Application telemetry • Distributed traces • Logs • Memory-dump analysis • WinDbg • Server diagnostics • Incident investigation

Observability & Application Performance: Dynatrace • AppDynamics • OpenSearch • Apdex • Performance analysis • Error-pattern analysis • Application health

Engineering & Automation: Python • PowerShell • JavaScript/TypeScript • SQL • Git/GitHub • Jenkins • Ansible • Playwright • Chrome DevTools

AI & Developer Tooling: Claude Code / CLI • Ollama • OpenAI API • MCP • Agentic workflows • Multi-agent orchestration • Deterministic validation

Systems & Application Development: Windows • Linux • .NET • Node.js • Angular • Svelte • Mainframe application analysis

PROFESSIONAL EXPERIENCE

Chubb Business Services India

Senior Site Reliability Engineer | Jan 2024 – Present

- Troubleshoot complex production issues using **Dynatrace telemetry, traces, logs, WinDbg, memory-dump analysis, browser/runtime evidence, and server-level diagnostics**, selecting the investigation approach based on the problem rather than relying on a single monitoring or troubleshooting tool.
- Design and develop custom engineering applications that automate recurring incident-resolution and operational workflows, reducing repetitive manual investigation and execution.
- Designed an **application-intelligence dashboard on top of Dynatrace data** to surface recurring error patterns and performance degradation. Developed a hierarchical performance model that evaluates underlying calls using Apdex and aggregates those signals by business functionality, allowing application areas to be ranked by performance rather than relying only on an application-wide score.
- Building and iterating on an **agentic application health-check approach** that exercises real web-application workflows using AI agents. The workflow can retrieve Dynatrace diagnostic information when available; for automation servers without Dynatrace instrumentation, developed a lightweight .NET metrics collector to provide essential server-health signals.

- Experiment with multiple **AI-assisted troubleshooting approaches**. A Smart Browser extension collects and synthesizes investigation context distributed across browser tabs, while a separate Claude CLI environment uses Chrome DevTools MCP, Playwright MCP, and Dynatrace access to reproduce behavior, inspect runtime/network evidence, and assist investigations.
- Continue to refine AI-assisted workflows based on limitations observed during use, including probabilistic or inconsistent summaries, excessive verbosity, and poor signal prioritization; move repeatable analysis toward deterministic processing where appropriate.
- Define engineering problem statements for internship programs across code intelligence, agentic UI, and AI-assisted observability, mentor engineers through the work, and remain hands-on in maintaining and evolving the resulting systems.
- Apply application-discovery and runtime-validation techniques to **mainframe modernization**, mapping TELON screens and fields to code/database relationships and using automated application execution and database evidence to validate inferred relationships.

Cognizant Technology Solutions (Client: Chubb)

Senior Software Engineer | Jul 2017 – Jan 2024

- Built automation, observability solutions, and engineering tooling using **Python, PowerShell, JavaScript**, and web technologies for a large enterprise insurance application.
- Used **AppDynamics, OpenSearch, application analytics, logs, and runtime evidence** to investigate application behavior, identify performance bottlenecks, and improve operational visibility.
- Developed custom tools and automation to reduce repetitive development, support, and operational work rather than depending solely on off-the-shelf tooling.
- Collaborated with DevOps teams on CI/CD workflows and used **Jenkins and Ansible** as part of application deployment and operational automation.

Cognizant Technology Solutions (Client: Chubb)

Senior Associate | Apr 2010 – Jul 2017

- Worked on enterprise insurance applications on IBM Mainframe and developed automation to reduce repetitive engineering work.
- In 2013, learned Python and moved into development of a custom web application, progressively expanding into web technologies and automation. This transition established the foundation for the production engineering, application troubleshooting, and observability work that followed.

Infosys

Technology Analyst | Jun 2006 – Apr 2010

- Developed and supported enterprise mainframe applications using **COBOL and DB2**.
- Created automation workflows to reduce repetitive development and operational tasks.
- Received the **Infosys Excellence Award** and **Prima Award – Project Execution** for contributions to project execution and delivery.

SELECTED ENGINEERING SYSTEMS

AI Software Engineering Harness

Designed and built a local-first multi-agent software engineering harness using **Ollama**, coordinating specialized agents for planning, implementation, review, testing, and verification. Added an independent execution and verification layer so generated implementations are validated against test cases rather than relying solely on model-reported outcomes.

Currently extending the approach with a containerized application environment for functional and regression validation during agentic coding sessions, addressing the gap between successful unit/build checks and actual enterprise-application behavior.

Code Intelligence

Developed approaches for understanding relationships across application repositories using code analysis and graph-based representations. Reworked an initially LLM-driven natural-language-to-Cypher approach after observing that differently worded versions of the same question could produce inconsistent queries, moving recurring information needs toward **deterministic graph queries**.

Application Discovery & Modernization

Building application-discovery and runtime-validation workflows for legacy/mainframe systems, connecting user-interface fields and screens with code and database relationships and validating inferred relationships through automated execution and resulting database changes.

TECHNICAL LEADERSHIP

- Define real engineering problem statements and mentor engineers while allowing them to independently investigate and develop solutions.
- Use internship programs to develop SRE engineers with stronger software-engineering and automation capabilities; members of the first internship cohort subsequently joined the team.
- Continue hands-on ownership after experimental projects conclude, including changing architecture when an approach proves inconsistent or unsuitable.
- Prefer a hands-on individual-contributor / technical-leadership path over people management.

EDUCATION

University of Pennsylvania — Master of Computer and Information Technology | 2021 – 2024

International Institute of Information Technology Hyderabad (IIIT-H) — Master of Science in Data Science (MSc DS) — In Progress

AWARDS

Infosys Excellence Award • Prima Award – Project Execution